OR4023 (Optimization) Homework 2

Name:_____ Student ID:_____

Question 1. Consider the problem of finding the point on the line $x_1 + x_2 = 2$ that is closest to the point $(2, 2)^T$.

a) Write the problem as a constrained optimization problem.

b) Convert the problem into an unconstrained optimization problem by eliminating x_2 from the objective function.

c) Solve the unconstrained optimization problem.

Answer.

a) Constrained optimization problem:

$$\begin{cases} \min f(x_1, x_2) = (x_1 - 2)^2 + (x_2 - 2)^2 \\ \text{s.t. } x_1 + x_2 = 2 \end{cases}$$

b) Eliminate the constraint by substituting $x_2 = 2 - x_1$ into the objective function:

$$\min g(x_1) = (x_1 - 2)^2 + x_1^2.$$

c) We have $g(x) = 2x^2 - 4x + 4$ and $g'(x) = 4x - 4 = 0$ implies $x = 1$. Since g is convex, $x = 1$ is the unique global minimizer.

Question 2. A cylindrical can is to be designed to hold a volume of $V = 500\,\mathrm{cm}^3$. The goal is to minimize the cost of material used to construct the can, which is proportional to its surface area. The radius r must be at least 2 cm and at most 10 cm . The height h must be at least 5 cm and at most 20 cm . Formulate this problem as a constrained nonlinear optimization problem.

Answer. The surface area function is

$$S(r, h) = 2\pi rh + 2\pi r^2 = 2\pi(rh + r^2).$$

Since 2π is a constant, we can omit it from the objective function. The constrained optimization problem is

$$\begin{aligned} & \min f(r,h) = rh + r^2 \\ \text{s.t.} \quad & \pi r^2 h = 500, \\ & 2 \leqslant r \leqslant 10, \\ & 5 \leqslant h \leqslant 20. \end{aligned}$$

Question 3. Suppose f is an unimodal function in $[a,b]$ with the minimum point x^*. Let l, r be such that $a < l < r < b$. Show

a) If $f(l) > f(r)$, then $x^* \in (l,b)$.

b) If $f(l) = f(r)$, then $x^* \in (l,r)$.

Answer. Since f is unimodal and x^* is the minimum, we know f is strictly increasing for $x > x^*$ and strictly decreasing for $x < x^*$. We use proof by contradiction.

a) If $x^* \notin (l,b)$, then $x^* \leqslant l < r$. Hence $f(l) < f(r)$. Contradiction.

b) If $x^* \leqslant l$, then as in (a), we have $f(l) < f(r)$. Contradiction. If $x^* \geqslant r$, then $l < r \leqslant x^*$ implies $f(l) > f(r)$. Contradiction. Hence $x^* \in (l,r)$.

Question 4. Let $f(x) = x^2 + e^{-x}, x \in [0,1]$.

a) Show that f is unimodal.

b) Apply the exhaustive search method.

i. Determine the required number of points so that the length of the final interval of uncertainty is bounded by 0.1.

ii. Apply the exhaustive search method to find the final interval of uncertainty.

c) Apply the golden section search method.

i. Determine the required number of iterations and the number of function evaluations so that the length of the final interval of uncertainty is bounded by 0.1.

ii. Apply the golden section search method to find the final interval of uncertainty.

d) Apply the bisection method.

i. Determine the required number of iterations and the number of function evaluations so that the length of the final interval of uncertainty is bounded by 0.1.

ii. Apply the bisection method to find the final interval of uncertainty.

iii. Write a computer program in MATLAB for the bisection method and verify your answer in part (ii).

Answer. Remark: the question is to find the minimum of f.

a) $f'(x) = 2x - e^{-x}$. Evaluate: $f'(0) = -1 < 0$, $f'(1) = 2 - e^{-1} > 0$. Since $f'(x)$ is continuous and shifts from negative to positive, it has a root in $[0, 1]$. Check $f''(x) = 2 + e^{-x} > 0$, so $f'(x)$ is strictly increasing, implying a single root $x = c$. Thus, $f(x)$ decreases to a minimum at c then increases, establishing unimodality.

b) If we divide the interval $[0, 1]$ into N equal subintervals with points

$$0 = x_0 < x_1 < x_2 < \cdots < x_N = 1,$$

and the minimum is attained at x_k, then the interval of uncertainy is $[x_{k-1}, x_{k+1}]$. We require its length $2/N \leqslant 0.1$, yielding $N \geqslant 20$. Take $N = 20$. The total number of points is 21 (including the end points).

Using a computer program, we compute $f(x_i), i = 0, 1, \ldots, 20$ and find the minimum is attained at $x_7 = 0.35$. Hence the interval of uncertainty is $[x_6, x_8] = [0.3, 0.4]$. Then length is 0.1.

c) Let $\tau = (\sqrt{5} - 1)/2$ and N be the number of iterations. The length of final interval of uncertainty is τ^N. Then $\tau^N \leqslant 0.1$ implies $N \geqslant \ln(0.1)/\ln(\tau) \approx 4.8$. Hence we take $N = 5$. The number of function evaluations is $N + 1 = 6$. Note that $\tau^2 = 1 - \tau$.

i. Take $x_s = 0, \hat{x} = \tau^2, \bar{x} = \tau, x_t = 1$. $f(\hat{x}) \approx \mathbf{0.8284} < f(\bar{x}) \approx \mathbf{0.9210}$. Discard $[\bar{x}, x_t]$.

The new interval is $[0, \tau]$ with length τ.

ii. Take $x_s = 0, \hat{x} = \tau^3, \bar{x} = \tau^2, x_t = \tau$. $f(\hat{x}) \approx \mathbf{0.8455} > f(\bar{x}) \approx 0.8284$. Note the new $\bar{x}$ is equal to the previous $\hat{x}$, so $f(\bar{x})$ need no recalculation. Discard $[x_s, \hat{x}]$.

The new interval is $[\tau^3, \tau]$ with length τ^2.

iii. Take $x_s = \tau^3, \hat{x} = \tau^2, \bar{x} = 2\tau^3, x_t = \tau$. $f(\hat{x}) \approx 0.8284 < f(\bar{x}) \approx \mathbf{0.8466}$. Note the new $\hat{x}$ is equal to the previous $\bar{x}$, so $f(\hat{x})$ need no recalculation.

Discard $[\bar{x}, x_t]$.

The new interval is $[\tau^3, 2\tau^3]$ with length τ^3.

iv. Take $x_s = \tau^3, \hat{x} = \tau^3 + \tau^5, \bar{x} = \tau^2, x_t = 2\tau^3$. $f(\hat{x}) \approx \mathbf{0.8281} < f(\bar{x}) \approx 0.8284$. Note the new $\bar{x}$ is equal to the previous $\hat{x}$, so $f(\bar{x})$ need no recalculation. Discard $[\bar{x}, x_t]$.

The new interval is $[\tau^3, \tau^2]$ with length τ^4.

v. Take $x_s = \tau^3, \hat{x} = \tau^3 + \tau^6, \bar{x} = \tau^3 + \tau^5, x_t = \tau^2$. $f(\hat{x}) \approx \mathbf{0.8321} > f(\bar{x}) \approx 0.8281$. Note the new $\bar{x}$ is equal to the previous $\hat{x}$, so $f(\bar{x})$ need no recalculation. Discard $[x_s, \hat{x}]$.

The final interval is $[\tau^3 + \tau^6, \tau^2]$, which is approximately $[0.2918, 0.3820]$. The length is 0.0902.

The 6 function evalutions are marked in bold faces.

d) After N iterations, the interval of uncertainty length is $2^{-N} \leqslant 0.1$ implies $N \geqslant \ln(10)/\ln(2) \approx 3.3$. So take $N = 4$. The number of function evaluations is $N + 2 = 6$.

We have $g(x) = f'(x) = 2x - e^{-x}$. Use bisection method to find the zero of g.

1. $a = 0, b = 0.5, c = 1$. $g(a) = -1 < 0$, $g(b) = 1 - e^{-0.5} > 0$, $g(c) = 2 - e^{-1} > 0$. New interval is $[0, 0.5]$.

2. $a = 0, b = 0.25, c = 0.5$. $g(b) < 0$. New interval $[0.25, 0.5]$.

3. $a = 0.25, b = 0.375, c = 0.5$. $g(b) > 0$. New interval $[0.25, 0.375]$.

4. $a = 0.25, b = 0.3125, c = 0.375$. $g(b) < 0$. New interval $[0.3125, 0.375]$. Length 0.0625.

MATLAB code:

```
a = 0; b = 1; tol = 0.1;
for i = 1:4
    c = (a + b)/2;
    g = 2*c - exp(-c);
    if g > 0
        b = c;
    else
        a = c;
    end
    fprintf('New interval: [%.4f, %.4f]\n', a, b);
end
```